

HOANG GIANG VU

PhD Candidate in Computer Science / Artificial Intelligence

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Academic Profile

PhD candidate in Computer Science at the University of Tours specializing in Knowledge Graphs, Multimodal Knowledge Graphs, NLP, and multimedia understanding. Research includes the construction of the largest publicly available French television knowledge graph, the development of a large-scale TV-video dataset for multimodal Named Entity Recognition (NER), knowledge-driven NER for French TV transcripts, and KG-enhanced retrieval and question answering. Experienced in scientific publication, student supervision, and international research collaboration.

Research Interests

Knowledge Graphs and Multimodal Knowledge Graphs; Natural Language Processing and Information Extraction; Retrieval-Augmented Generation; Speech and Video Understanding.

Education

PhD in Computer Science

Nov 2023 – Present

University of Tours, Tours, France

- Dissertation: *Video Description by Textual and Semantic Enrichment*
- Supervisors: Prof. Mathieu Delalandre, Prof. Nathalie Friburger, and Prof. Arnaud Soulet
- Expected defense: January 2027

Master's Degree in Engineering and Computer Science

Sep 2019 – Oct 2022

University of Messina, Messina, Italy

Bachelor's Degree in Information and Technology

Aug 2014 – Jun 2018

University of Transport and Communications, Hanoi, Vietnam

Teaching Experience

Teaching Assistant

Oct 2022 – Mar 2023

University of Messina, Messina, Italy

- Supported teaching and tutoring activities for the Algorithms and Data Structures course.
- Provided support in basic Python programming.

Teaching Related Role During PhD

2024 - Present

University of Tours

- Supervised and mentored five groups of undergraduate and Master's students on applied research projects in Natural Language Processing, speech processing, and multimedia analysis.
- Guided student projects on NER tasks, including rule-based approaches using CasEN, BERT-based models, and speech-to-text processing of TV content using OpenAI Whisper.
- Supported the development and optimization of Whisper-based speech recognition pipelines, including parallel processing techniques to accelerate transcription of large-scale TV video collections.
- Contributed to student projects involving the capture and analysis of sign language from television content for accessibility-oriented applications supporting people with disabilities.
- Participated in student supervision within both French and international collaborations, including collaborations with Hanoi University of Science and Technology (HUST) and Hong Duc University

(HUD), Vietnam.

Research Experience

PhD Researcher in Multimodal Knowledge Graphs

2023 – Present

LIFAT, University of Tours, Tours, France

- Conducting research at the intersection of Knowledge Graphs, Multimodal Knowledge Graphs, Natural Language Processing, and multimedia content understanding, with a particular focus on the television domain.
- Designed and constructed stvd-kg, a large-scale knowledge graph derived from French Electronic Program Guide (EPG) data, integrating approximately 14M entities and 65M triples.
- Proposed a knowledge-driven Named Entity Recognition framework for French TV video transcripts, combining transcript-based processing, external knowledge, and a multi-level speech recognition strategy.
- Constructed a large-scale TV-video dataset for multimodal Named Entity Recognition, comprising approximately 8,000 hours of video, 4,000 unique entities, and more than 126,000 entity occurrences.
- Developed a Retrieval-Augmented Generation (RAG) demonstrator integrating stvd-kg for knowledge-grounded retrieval and question answering over television content.
- Conducted a systematic study of Knowledge Graph and Multimodal Knowledge Graph technologies for television, identifying limitations in existing multimedia ontologies, multimodal knowledge representation methods, and KG construction technologies for the TV domain.
- Authored scientific publications and presented research on knowledge-driven NER, knowledge graphs, multimodal knowledge representation, and intelligent access to multimedia archives.

Research Assistant

Sep 2021 – May 2022

University of Messina, Messina, Italy

Research activities focused on Computer Vision and anomaly detection.

- Worked on steel defect detection using computer-vision and deep-learning techniques.
- Investigated predictive-maintenance methods using CNN, LSTM, and attention mechanisms on the NASA C-MAPSS dataset.

Publications

Peer-Reviewed International Conference Papers

1. **H. G. Vu**, M. Delalandre, N. Friburger, and A. Soulet, “Knowledge-Driven Named Entity Recognition in French TV Video Transcripts,” accepted for publication in the *2026 IEEE International Conference on Content-Based Multimedia Indexing (CBMI 2026)*, Toulouse, France, 2026. *Forthcoming*.
2. **H. G. Vu**, N. Friburger, A. Soulet, and M. Delalandre, “stvd-kg: A Knowledge Graph for French Electronical Program Guides,” in *Web Information Systems Engineering – WISE 2025*, Lecture Notes in Computer Science, vol. 16367, pp. 390–405, Springer, 2026. DOI: [10.1007/978-981-95-7394-3_27](https://doi.org/10.1007/978-981-95-7394-3_27).

Journal Manuscript in Preparation

1. **H. G. Vu**, N. Friburger, A. Soulet, and M. Delalandre, “Knowledge Graph and Multimodal Knowledge Graph Construction for Television Multimedia: A Survey,” *Manuscript in preparation*.

Conference Presentations

- **Oral Presenter**, “Knowledge-Driven Named Entity Recognition in French TV Video Transcripts,” *2026 IEEE International Conference on Content-Based Multimedia Indexing (CBMI 2026)*, Toulouse, France, October 2026.

- **Oral Presenter**, “stvd-kg: A Knowledge Graph for French Electronical Program Guides,” *26th International Conference on Web Information Systems Engineering (WISE 2025)*, Marrakech, Morocco, December 2025.
- **Poster Presenter**, “Knowledge Graph Applications in the Mental Health Domain,” *LE STUDIUM Conference: Early-Life Adversity and Resilience: Multi-Disciplinary Perspectives in a Rapidly Changing World*, Tours, France, 3–5 September 2024.

Industry Experience

Data Analyst	Nov 2021 – Mar 2022
<i>NIBELC Germany GmbH, Berlin, Germany</i>	
• Conducted labour-market analysis in Ireland, Hungary, Poland, and Slovenia. • Collected web data using Scrapy. • Processed and analysed datasets using Python, NumPy, and Pandas. • Organised labour-market data by region, company, and workforce characteristics. • Visualised data using Elasticsearch and Kibana. • Contributed to business-development activities involving European workforce companies including EWL Group and Otto Workforce Group.	

Technical Skills

Programming & Data	Python, C++, SPARQL, NumPy, Pandas
Deep Learning	Transfer Learning, Named Entity Recognition, Entity Linking, Retrieval-Augmented Generation
Knowledge Representation	Knowledge Graphs, Multimodal Knowledge Graphs, Semantic Technologies
Research Tools	Git, GitHub, Jupyter, GraphDB, LaTeX, Scrapy

Scholarships and Awards

Fully Funded Scholarship	2019 – 2020
<i>University of Messina</i>	
Fully Funded Scholarship	2020 – 2022
<i>ERSU Messina</i>	
Fully Funded PhD Position	2023 – 2026
<i>LIFAT – University of Tours, France</i>	

Languages

Vietnamese	Native
English	Professional working proficiency; graduate studies, research, and professional activities conducted in English. IELTS Academic 6.5 (2018).